Goal: Interpret the definite integral as the (signed) area underneath the graph of a function.
Recall Left/Right Riemann sums (L_N and R_N) are used to approximate the (signed) area underneath the graph of a function.

Question: What happens when we take $N \to \infty$? See what happens with a Desmos presentation (link here).

$$\lim_{N\to\infty} L_N = \lim_{N\to\infty} R_n = \text{The exact area}$$

Definition. The **(definite) integral** of a function f from a to b is denoted as follows:

$$\int_a^b f(x)\,dx \overset{\text{def}}{=} \lim_{N\to\infty} L_N = \lim_{N\to\infty} R_n$$

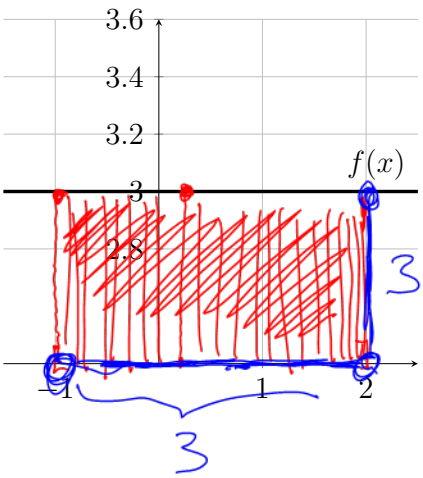
(b: Upper bound; a: Lower Bound)

$=$ The signed area between the graph of f and the x-axis on $[a,b]$.

a and b are called the upper and lower limits (or bounds) of integration, respectively. If the integral is finite, we say f is (Riemann) integrable on $[a,b]$.

"The integral of 3 from -1 to 2"

Example 1. Compute $\int_{-1}^{2} 3\,dx$

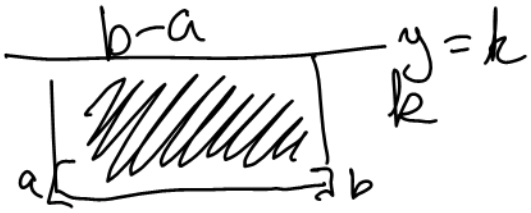


$$\int_{-1}^{2} 3\,dx = \text{Area of the Rectangle}$$

$$= [\text{Base}]\,[\text{Height}]$$

$$= 3(3)$$

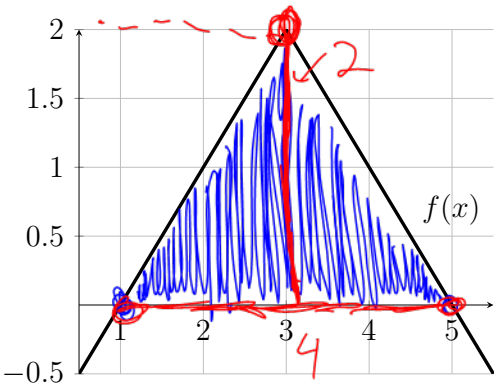
$$= 9$$



NOTE: In general, if k is a constant:

$$\int_a^b k\,dx = k(b-a)$$

Example 2. Compute $\int_1^5 (2-|x-3|)\,dx$

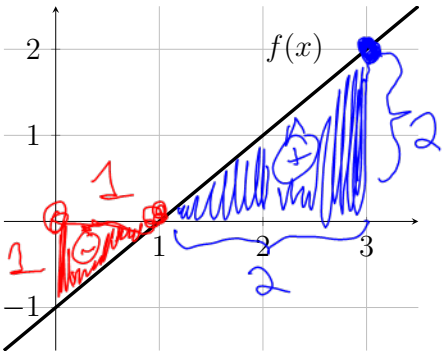


$\int_1^5 (2-|x-3|)\,dx$ = Area of Triangle

$= \frac{1}{2}$[Base][Height]

$= \frac{1}{2}[4][2] = 4$

Example 3. Compute $\int_0^3 (x-1)\,dx$



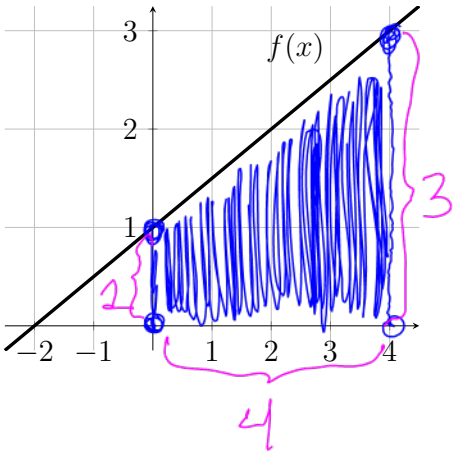
$\int_0^3 (x-1)\,dx$ = Area of Yellow Triangle − Area of Blue Triangle

$= \frac{1}{2}(2)(2) - \frac{1}{2}(1)(1)$

$= 2 - \frac{1}{2} = \frac{3}{2}$ $+\frac{1}{2}(1)(-1)$

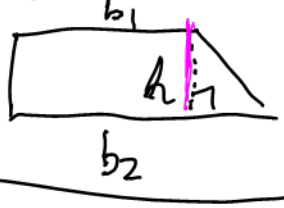
Example 4. Compute $\int_0^4 \left(\frac{1}{2}x+1\right)\,dx$

(a) METHOD 1:



Area of a Trapezoid:

$A = \frac{1}{2}(b_1+b_2)h$

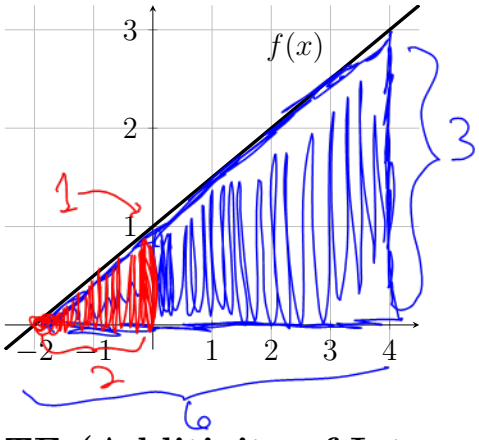


$\int_0^4 \left(\frac{1}{2}x+1\right)dx$ = Area of Trapezoid

$= \frac{1}{2}(1+3)(4)$

$= 8$

(b) METHOD 2:



$\int_0^4 \left(\frac{1}{2}x+1\right) dx =$ Area of Yellow Triangle $-$ Area of Blue Triangle

$= \frac{1}{2}(6)(3) - \frac{1}{2}(2)(1)$

$= 9 - 1$

$= 8$

NOTE (Additivity of Integrals Example):

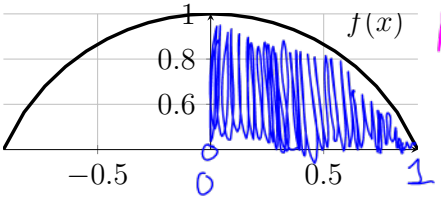
Rewriting METHOD 2 using integrals, we get:

$$\int_0^4 \left(\frac{1}{2}x+1\right) dx = \int_{-2}^{4} \left(\frac{1}{2}x+1\right) dx - \int_{-2}^{0} \left(\frac{1}{2}x+1\right) dx$$

I.e.,

$$\int_{-2}^4 \left(\frac{1}{2}x+1\right) dx = \int_{-2}^{0} \left(\frac{1}{2}x+1\right) dx + \int_{0}^{4} \left(\frac{1}{2}x+1\right) dx$$

Example 5. Compute $\int_0^1 \sqrt{1-x^2}\, dx$



$y = \sqrt{1-x^2}$

$y^2 = 1 - x^2$

$x^2 + y^2 = 1$

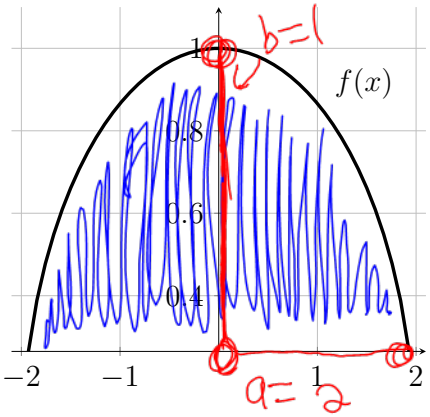
R^2

$\int_0^1 \sqrt{1-x^2}\, dx$

$= \frac{1}{4}$ [Area of Circ. w/ Rad. 1]

$= \frac{1}{4}\pi(1)^2 = \frac{\pi}{4}$

Example 6. Compute $\int_{-2}^{2} \sqrt{1-\frac{x^2}{2^2}}\, dx$



$y = \sqrt{1-\frac{x^2}{2^2}}$

$y^2 = 1 - \frac{x^2}{2^2}$

$\frac{x^2}{2^2} + \frac{y^2}{1^2} = 1$

$\int_{-2}^{2} \sqrt{1-\frac{x^2}{2^2}}\, dx$

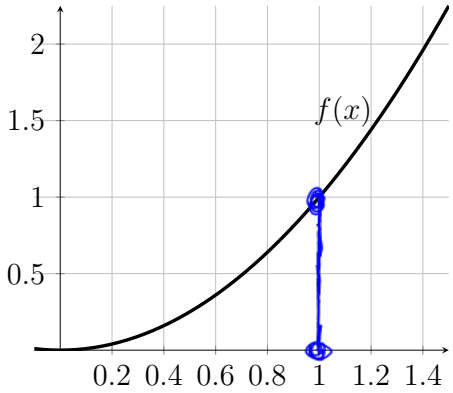
$= \frac{1}{2}$ [Area of the Ellipse, $a=2$, $b=1$]

$= \frac{1}{2}\pi(2)(1)$

$= \pi$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow$ Area is πab

Example 7. Compute $\int_1^1 x^2\ dx$



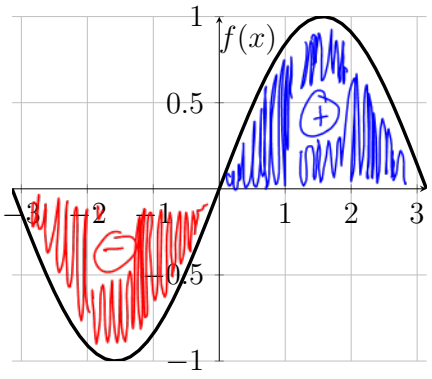
$\int_1^1 x^2\,dx$ = Area of the Line Segment

$= 0$

NOTE: In general,

$$\int_a^a f(x) = 0$$

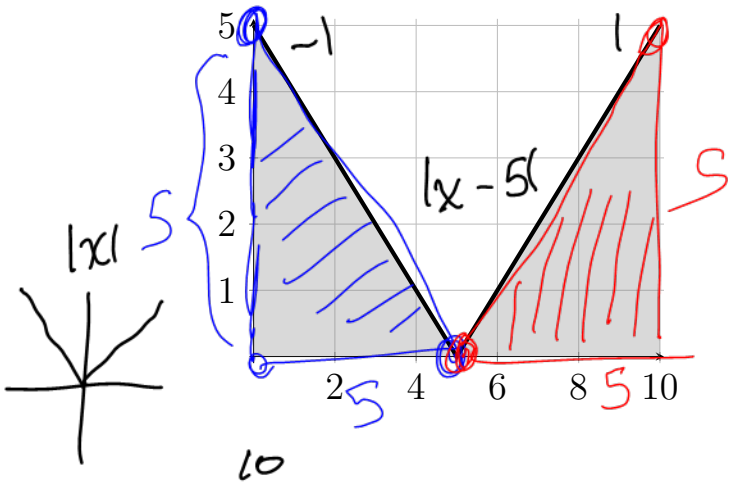
Example 8. Compute $\int_{-\pi}^{\pi} \sin x\ dx$



Same area, but one is + and one is −

$$\int_{-\pi}^{\pi} \sin x\,dx = 0$$

Example 9. Consider the shaded region below:



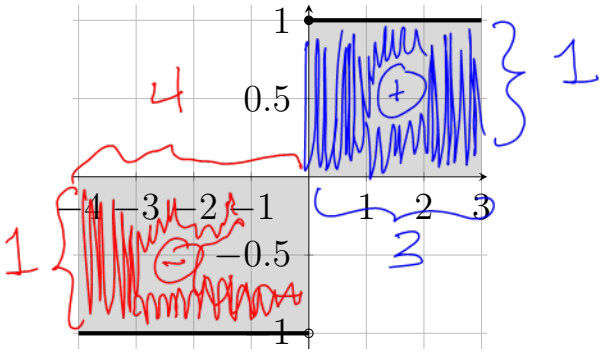
(a) Describe the area of the shaded region using an integral:

$$\int_0^{10} |x-5|\,dx$$

(b) Compute the area of the shaded region.

$$\int_0^{10} |x-5|\,dx = \frac{1}{2}(5)(5) + \frac{1}{2}(5)(5) = 25$$

Example 10. Consider the shaded region below:



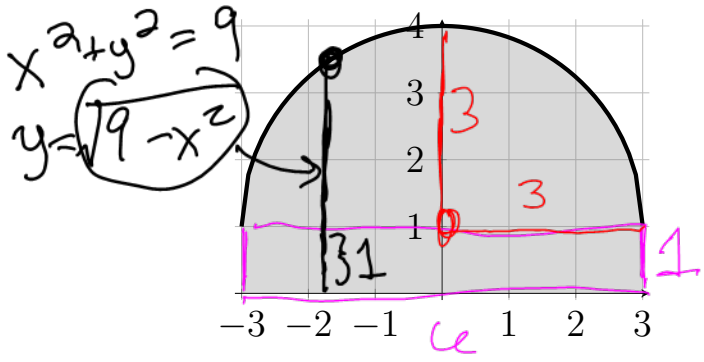
(a) Describe the area of the shaded region using an integral:

$$\int_{-4}^{3} f(x)\,dx \quad \text{where } f(x) = \begin{cases} -1 & x<0 \\ 1 & x\geq 0 \end{cases}$$

(b) Compute the area of the shaded region.

$$\int_{-4}^{3} f(x)\,dx = 3(1) - 4(1) = -1$$

Example 11. Consider the shaded region below:



(a) Describe the area of the shaded region using an integral:

$$\int_{-3}^{3} \left(1 + \sqrt{9-x^2}\right) dx$$

(b) Compute the area of the shaded region.

$$\int_{-3}^{3}\left(1+\sqrt{9-x^2}\right)dx = 6 + \frac{1}{2}\pi(3)^2$$
$$= 6 + \frac{9}{2}\pi$$

NOTE (Linearity of Integrals Example):

Rewriting our work using integrals, we see:

$$\int_{-3}^{3}(1+\sqrt{9-x^2})\,dx = \int_{-3}^{3}1\,dx + \int_{-3}^{3}\sqrt{9-x^2}\,dx$$